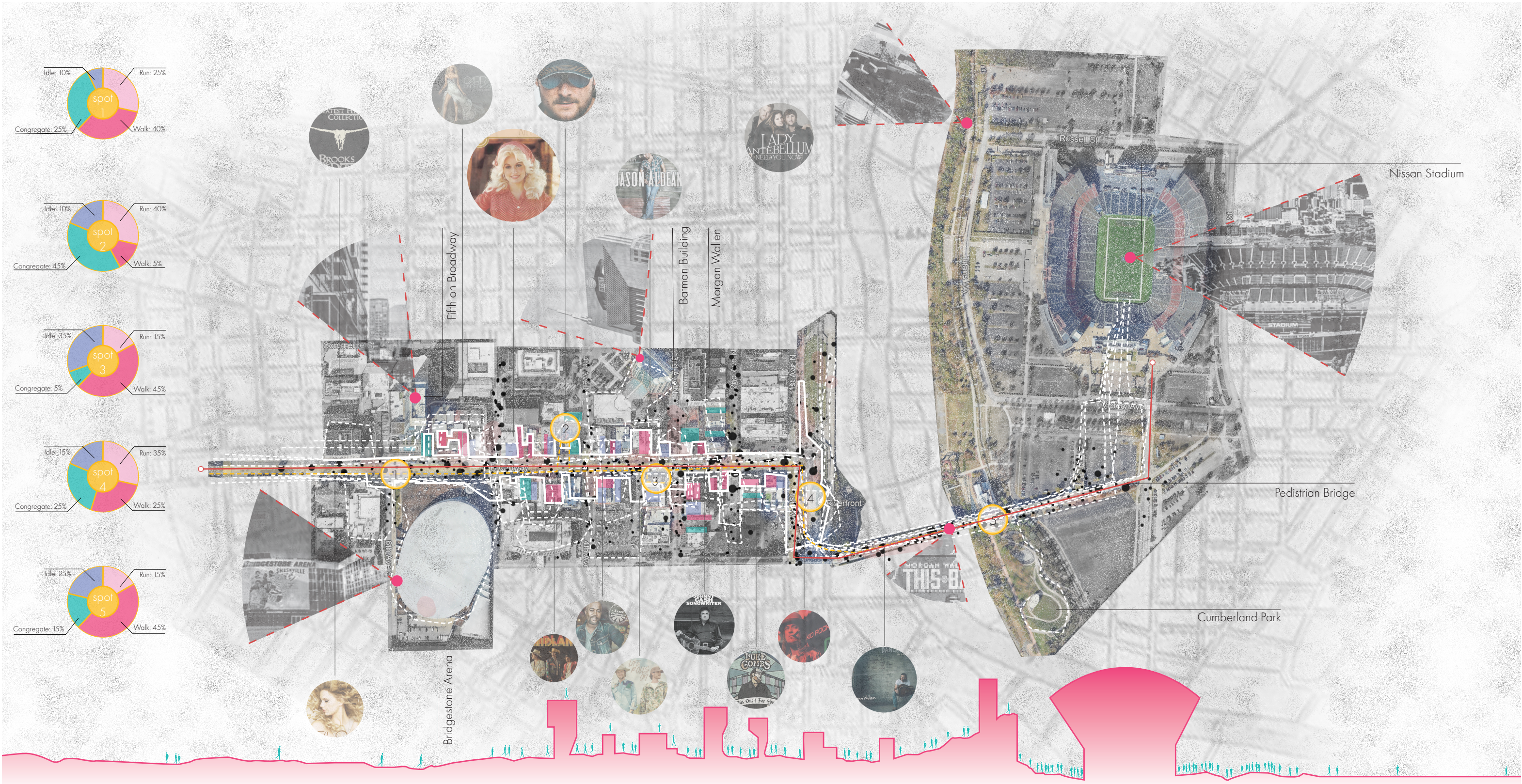




01. Walking Down Broadway (virtual immersive experience)

Class: 15-112
Instructor: Professor Mike Taylor
Partners: Individual Project

This project simulates the cultural fabric of Nashville through a first-person interactive game that blends **narrative design, real-time AI dialogue, and procedural content generation**. Players explore a dynamic, music-filled scene and engage in natural conversations with **AI-driven characters via the OpenAI API**. The system supports branching dialogue, contextual

awareness, and ambient soundscapes, creating a responsive, open-ended experience. The urban environment is animated with real-time **graphics, lighting simulations, and event-driven mechanics**. Every element—from shifting street energy to layered live music—is algorithmically choreographed to reflect the city's personality. Built entirely in **Python**, the project

uses **OpenCV** for **computer vision** and the **CMU Graphics Library** for **2D animation**. It demonstrates **object-oriented design, modular architecture, and real-time state management**. A **Markov chain**-based dialogue engine generates coherent, varied responses to simulate spontaneous conversation. Scene behaviors are governed by finite state machines, showcasing

computational thinking in orchestrating emergent behaviors. The project reflects a strong grasp of computational design, where **storytelling, interaction, and code structure converge to create a technically rich and emotionally engaging system**.

Please experience my game:
<https://www.youtube.com/watch?v=h1ZbFu-W11s>